Numerical Finance Project: Pricing of Bermudan Basket Options

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Introduction

This report presents the implementation of several variance reduction techniques for the pricing of European and Bermudan Call and Put option on a basket of stocks. The techniques implemented are the following:

- Antithetic Variables.
- Static Control Variate.
- Quasi Monte Carlo using Van der Corput and Halton Sequences.

Part 1: Code Review and Explication

Project Architecture

In this section we aim at briefly presenting the code architecture. We assume that the reader is already familiar with the part of code responsible for the generation of Random Numbers and processes.

- **Underlying**: An underlying is one of the base classes in the project. It is built on top the existing RandomProcess class. The main objective of this class is to be able to reconstruct the trajectories of a given underlying, given the diffusion process of the components of a universe. The features of this components, which shall be used for the diffusion will be defined in the RandomProcess class, when instantiating a process, such as BSEulerND or BSEulerNDAnti. Once again, in this context, we defined a class called **Basket** which build the value of the basket at each time step t as follows:

$$X_t = \sum_{i=1}^{d} \alpha_i S_t^i$$

  where X_t denotes the basket value at time t, α_i are the weights of each components i in the basket and S_t^i the values of each stock at time t.

- **Payoff**: This class is analogous to the R1R1 and R2R1 functions developed to solve the PDE in the course. It acts as a wrapper under which the user can derive a class and compute any payoffs, given a vector of prices. This class is meant to allow for the pricing of path dependent and strike forward options. It has been designed to be able to price that widest range of options possible This can then be used to compute the value of the payoff for any vector of prices. In the context of this program, we derived two child class allowing for the pricing of a call and put. Each take the last element of the vector to compute the respective payoffs.

- **Pricer**: Pricers are the main class of the project. The class primary inputs are the underlying on which the products will be priced and the rate which shall be used to discount the prices. The class **Payoff** is called in the price method, and is used to evaluate the option in a path-wise manner. The main method, Price() is a virtual method which is overriden in the child classes which are respectively **Monte Carlo** and **Longstaff Schwarz**.

  - **Monte Carlo**: This class evaluates options by taking the average of the terminal payoffs of each options. It is therefore only used for the pricing of European options. The price method takes an argument to decide whether the Control Variate method shall be implemented (on top of the other method which are chosen directly at the RandomProcess level).

- **Longstaff Schwarz**:
  - **LSStandard**: This is an implementation of the Longstaff Schwarz Algorithms where the function used for the regression is defined as follows:

$$Y_j = \sum_{l=1}^{L} X_t^l \alpha_l \tag{1}$$

    Please note that for questions of invertibility, and numerical stability, it is recommended not to used orders higher than 5. A default value of 3 has been set, and yields satisfying results.
  - **LSLaguerre**: This is another implementation of the Longstaff Schwarz Algorithms where we used the functions presented in the paper, known as the Laguerre polynomials. In the paper, it is mentioned that first 3 polynomials, scaled by an exponential factor yield decent results, as we shall see later. The polynomials used here are then the following

$$L_0(X) = \exp(-X/2), \tag{2}$$
$$L_1(X) = \exp(-X/2)\,(1 - X), \tag{3}$$
$$L_2(X) = \exp(-X/2)\,(1 - 2X + X^2/2), \tag{4}$$

    Note: Because of the exponentially decreasing factor, it is important that the user inputs prices of 1 for the spots (and scales the strike accordingly), otherwise the values of the polynomials decrease very rapidly to 0, leading to numerical inaccuracies. The prices returned by the price function are multiplied by 100 to rescale the price to the real value of the underlying.

Note: A map of the whole project is made available in the Appendix.

Pricer file

The file `pricer.cpp` is a simulation program used to price European and Bermudan Call and Put options using Monte Carlo methods, with support for variance reduction techniques and American-style pricing via Longstaff-Schwartz regression. The user can select among different pricing methods, such as:

- Standard Monte Carlo
- Monte Carlo with Antithetic Variables
- Monte Carlo with Control Variate
- Monte Carlo with Quasi-Random Numbers (Halton)
- Longstaff-Schwartz (standard or Laguerre polynomial basis) with variance reduction
- Black-Scholes analytical formula (for one-dimensional vanilla options)

All the variance reduction methodologies can be used simultaneously to optimize your results.

Parameter Configuration

Parameters are configured directly in the `main()` function at the top of the file. Below is a description of the key settings:

Option and Simulation Parameters:

- `strike = 100.0` // Strike price of the option
- `payoffType = "Call"` // "Call" or "Put"
- `startTime = 0.0, endTime = 1.0` // Maturity interval
- `nbSteps = 252` // Time discretization steps
- `nbSim = 10000` // Number of Monte Carlo paths

Underlying Parameters:

- `vecSpots = {100, 100, 100}` // Spot prices
- `vecRates = {0.05, 0.05, 0.05}` // Constant interest rate
- `vecWeights = {1, 0, 0}` // Basket weights
- `matCov = Inp->CSV2Mat("Inputs/matCov.csv")` // Covariance matrix (CSV input)

Method Toggles: Enable or disable specific pricing methodologies by setting the following booleans:

- `useMC` – Enable standard Monte Carlo
- `useAntithetic` – Enable Antithetic Variables
- `useControlVariate` – Enable Control Variate
- `useQuasi` – Enable Quasi-Monte Carlo (Halton)
- `useLS` – Enable Longstaff-Schwartz (standard basis)
- `useLSLaguerrePoly` – Enable Longstaff-Schwartz with Laguerre polynomials
- `useBS` – Use Black-Scholes analytical formula

Longstaff-Schwartz Configuration:

- `vecTimes = {0.2, 0.4, 0.6, 1.0}` // Possible exercise dates
- `orderLS = 3` // Polynomial order (for the standard regression basis)

Output Interpretation

After execution, the following information is displayed adapted to what you choose.

```
You priced a Call, Strike: 100, Maturity: 1Y.
You choosed the Monte Carlo + Antithetic Variable + Control Variate methodology.
Option price: 10.823
```

Compilation and Execution

To compile and run the program, use a command such as:

```
make pricer
./pricer
```

Part 2: Monte Carlo Pricer

European Call Option on mono underlying price and variance evolution

In this part, we will present the results we obtained with a Monte Carlo pricer. We first tried to prove that our base model is converging to the right value. To validate this convergence, we compared our numerical results with the analytical solution provided by the Black-Scholes closed-form formula. This step ensures that our Monte Carlo engine is correctly implemented and serves as a solid foundation for later variance reduction experiments. To do so, we evaluated a Call option on a basket of one underlying. In parallel, we verified whether the implemented variance reduction techniques — Antithetic Variables and Control Variate — also converge towards the correct theoretical value, thereby confirming both their correctness and usefulness.

- Antithetic Variables
- Control Variate

The following graphic highlights the evolution of the average option price with a different total number of simulations.

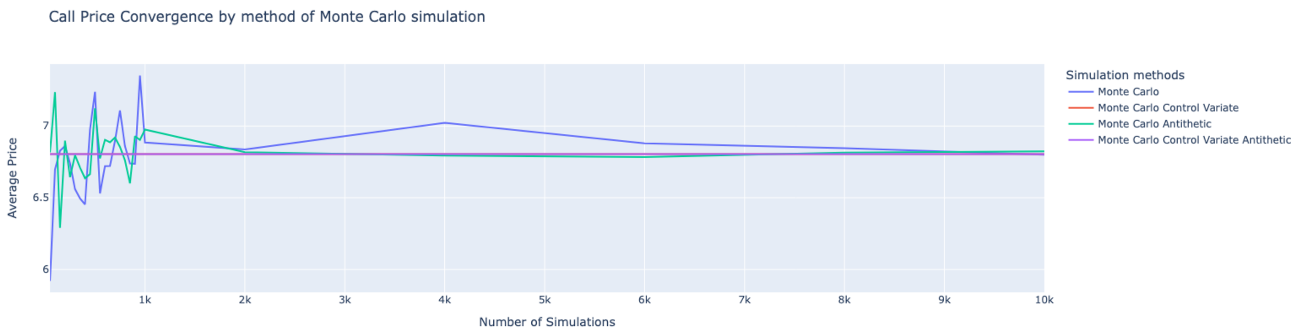

Figure 1: Average Call price evolution on a mono underlying basket

In order to highlight the variance gain of the difference methodology, we generated the following graphic plotting the evolution of the variance of the price estimator:

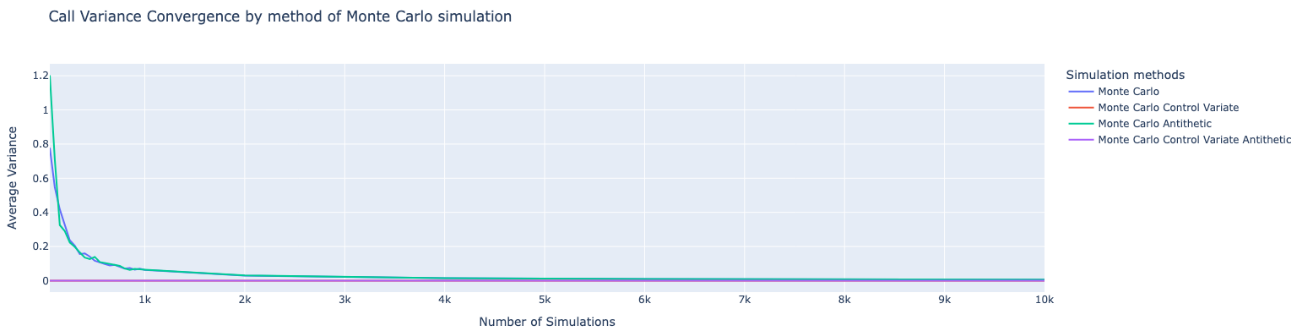

Figure 2: Average Call variance evolution estimator on a mono underlying basket

Based on the results generated by the simulations, we managed to summarize the results in the following table. One can remark that the confidence interval of the Monte Carlo simulation with the antithetic variable method is much closer to the real price of the option.

	MC	MC + CV	MC + Antithetic	MC + Antithetic + CV
Price	7.108469	6.80496	6.85432	6.80496
Variance	0.08154	0.0	0.086928	0.0
CI	[6.9487 ; 7.2683]	[6.8050 ; 6.8050]	[6.6839 ; 7.0247]	[6.8050 ; 6.8050]

Table 1: Monte Carlo variance reduction techniques comparison for a Call option on a mono underlying basket

European Put Option on mono underlying price and variance evolution

We then did the same reasoning for a Put option with the same parameters. We observed that the average option price also converged to the Black-Scholes price.

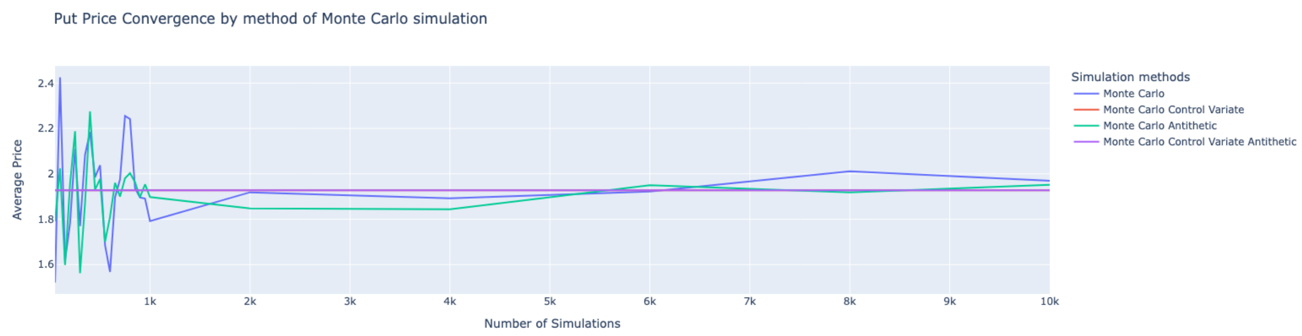

Figure 3: Average Put price evolution on a mono underlying basket

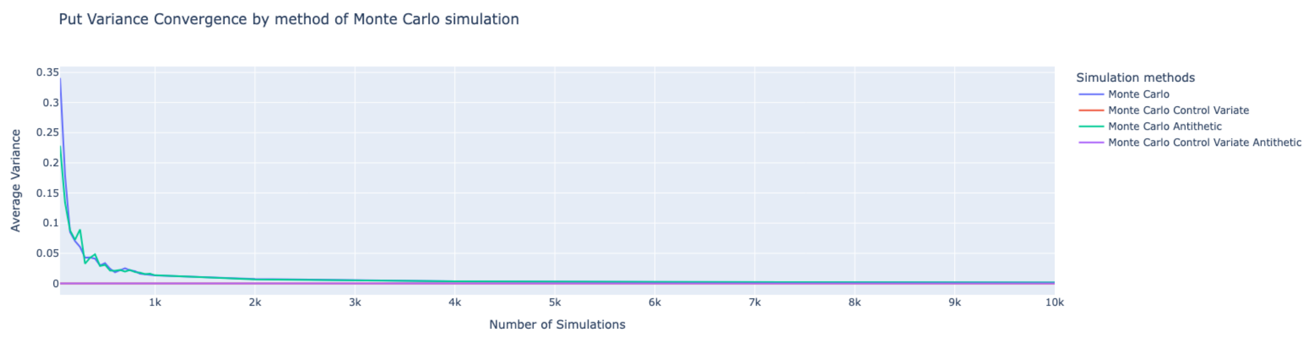

Figure 4: Average Put Variance evolution estimator on a mono underlying basket

Similar to the Call case, the convergence of the Put option price demonstrates the robustness of our Monte Carlo implementation. All methods approximate the Black-Scholes price within their respective confidence intervals, with variance reduction techniques significantly narrowing these intervals and thus increasing the reliability of the estimates.

	MC	MC + CV	MC + Antithetic	MC + Antithetic + CV
Price	2.255182	1.9279	1.980701	1.9279
Variance	0.021813	0.0	0.022397	0.0
CI	[2.2124 ; 2.2979]	[1.9279 ; 1.9279]	[1.9368 ; 2.0246]	[1.9279 ; 1.9279]

Table 2: Monte Carlo variance reduction techniques comparison for a Put option on a mono underlying basket

Once again, the antithetic plausible method forced the confidence interval to be closer to the desired value.

European Call Option on multi-underlying price and variance evolution

Now, our results have coherence in the values and also in the variance evolution. We tried to price options on a multi-underlying basket. We followed the same reasoning.

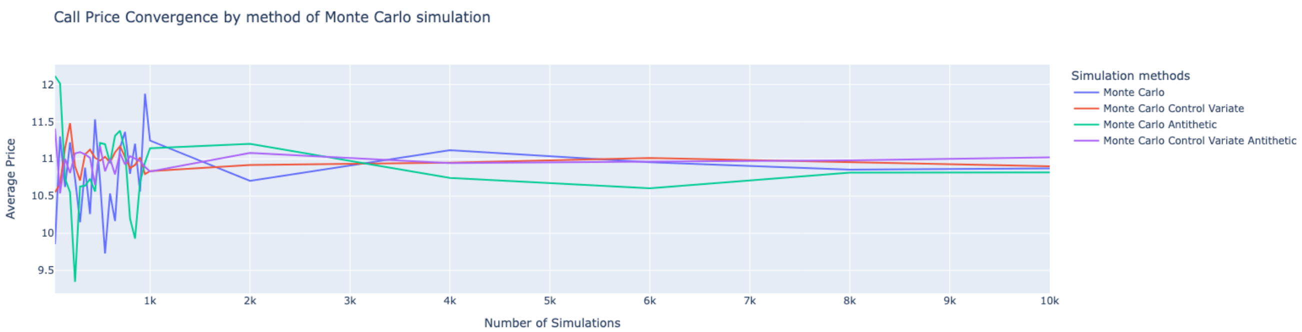

Figure 5: Average Call price evolution on a multi-underlying basket

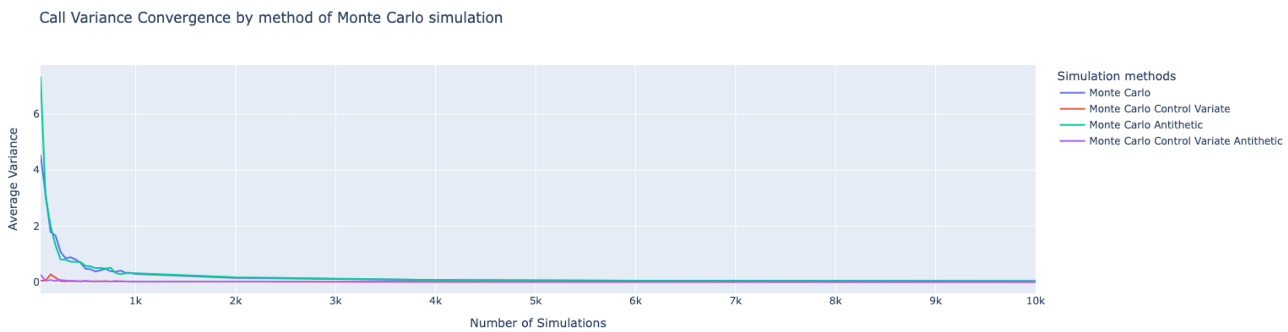

Figure 6: Average Call Variance evolution estimator on a multi-underlying basket

Figure 6 clearly illustrates that variance reduction techniques significantly stabilize the variance estimates across simulations. The naive Monte Carlo method exhibits higher fluctuations, while the combined Antithetic and Control Variate approach leads to a much smoother and lower variance curve, suggesting faster and more reliable convergence.

	MC	MC + CV	MC + Antithetic	MC + Antithetic + CV
Price	11.36181	11.025162	11.112208	10.92375
Variance	0.383006	0.014727	0.50762	0.012899
CI	[10.6111 ; 12.1125]	[10.9963 ; 11.0540]	[10.1173 ; 12.1071]	[10.8985 ; 10.9490]

Table 3: Monte Carlo variance reduction techniques comparison for Basket Call Option

Table 3 quantitatively confirms the observations made in the figures. The standard Monte Carlo method results in a relatively high variance and wide confidence interval. The Control Variate technique alone brings a drastic variance drop, and when combined with Antithetic sampling, it yields both the lowest variance and the narrowest confidence interval. Interestingly, all methods converge to similar price estimates, validating the correctness of our implementations.

This experiment highlights that, in the context of basket options where the payoff depends on the joint behavior of several underlying assets, variance reduction techniques are essential. They drastically improve convergence speed and result stability, making them highly recommended for pricing such multi-asset derivatives.

European Put Option on multi-underlying price and variance evolution

We extended the analysis to European Put options written on a basket of assets. As before, we compared the variance behavior and convergence speed of various Monte Carlo estimators.

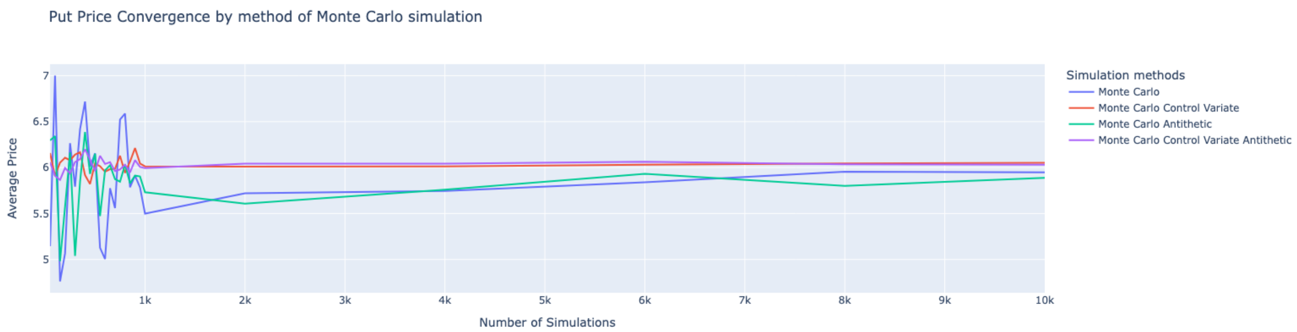

Figure 7: Average Put price evolution on a multi-underlying basket

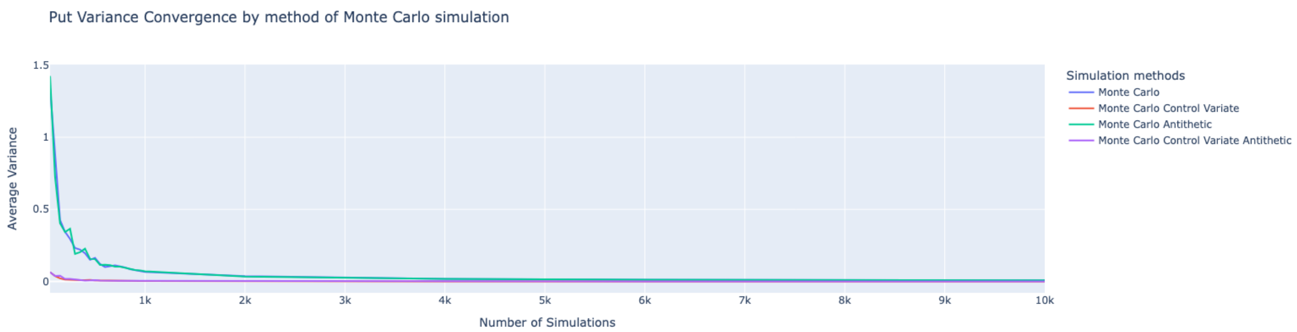

Figure 8: Average Put Variance evolution estimator on a multi-underlying basket

	MC	MC + CV	MC + Antithetic	MC + Antithetic + CV
Price	6.522318	6.128385	5.845686	5.979901
Variance	0.104099	0.004519	0.102492	0.005319
CI	[6.3183 ; 6.7264]	[6.1195 ; 6.1372]	[5.6448 ; 6.0466]	[5.9695 ; 5.9903]

Table 4: Monte Carlo variance reduction techniques comparison for Basket Put Option

The results mirror those obtained with the Call basket: Control Variate and Antithetic techniques each reduce variance significantly, but their combination offers the best performance. The variance falls by over 95% compared to the naive Monte Carlo method, and the confidence intervals become narrower and more centered around the true price.

European Call Option on multi-underlying variance evolution with Quasi MC

For the option variance evolution, we studied the variance of the price estimator with the Quasi Monte Carlo method and obtained the following results for both options

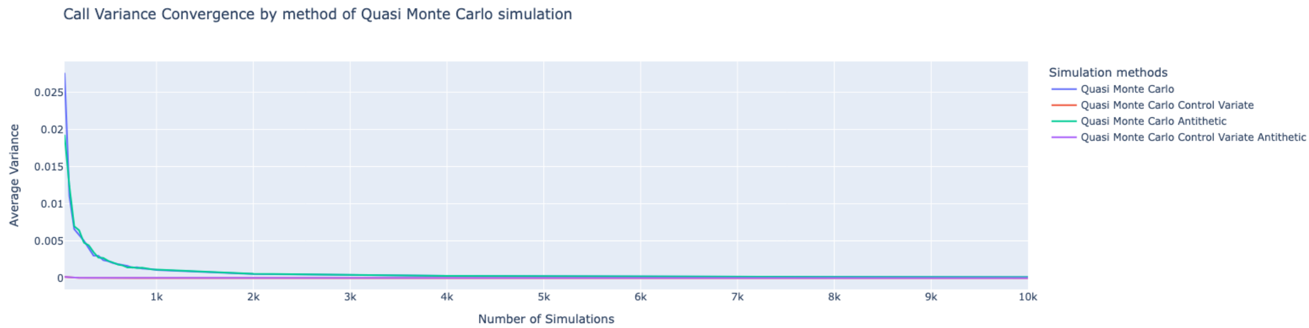

Figure 9: Average Call Variance evolution on a multi-underlying basket using Quasi MC

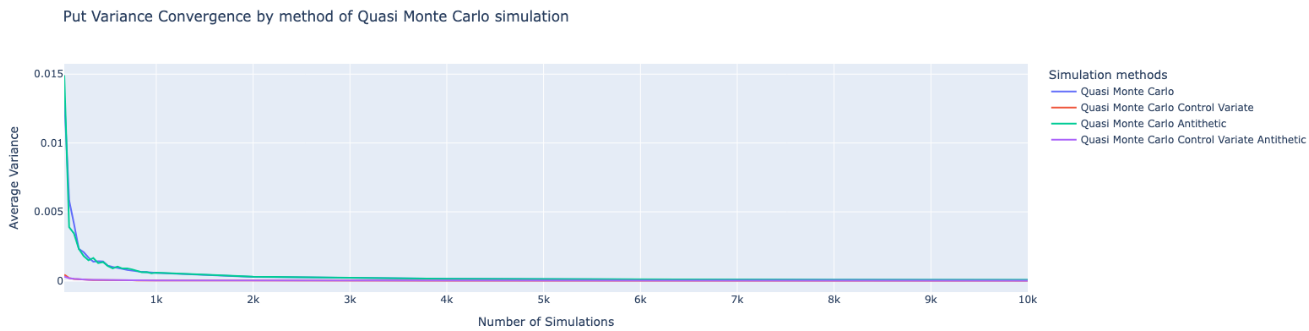

Figure 10: Average Put Variance evolution on a multi-underlying basket using Quasi MC

The use of Quasi-Monte Carlo methods (based on low-discrepancy sequences such as Halton or Sobol) further enhances variance reduction. As shown in Figures 9 and 10, the variance of the price estimator decreases faster than with standard random sampling. This confirms the efficiency of quasi-random sequences in high-dimensional integration tasks such as basket option pricing.

Part 3: Longstaff Schwarz Pricer

In this final part of the project, we aim at evaluating Bermudan options. Let us recall that these are options which can be exercised on a discrete set of dates.

From now on, we shall refer to the standard regression algorithm as LS, and the algorithms using Laguerre polynomials as LS Laguerre.

Let us now investigate our empirical results.

Convergence to Monte Carlo Prices for European Options

In order to evaluate our algorithm, let us first ensure that we match the prices presented above for the European option on both a mono underlying and a basket.

The figures below present the results of the simulations using both LS and LS Laguerre algorithms.

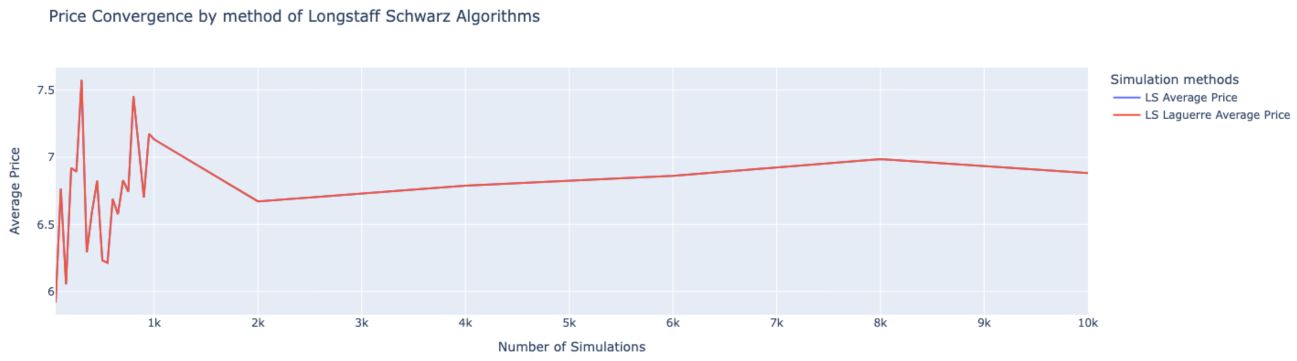

Figure 11: Longstaff Schwarz Pricing for European Option on Mono Underlying

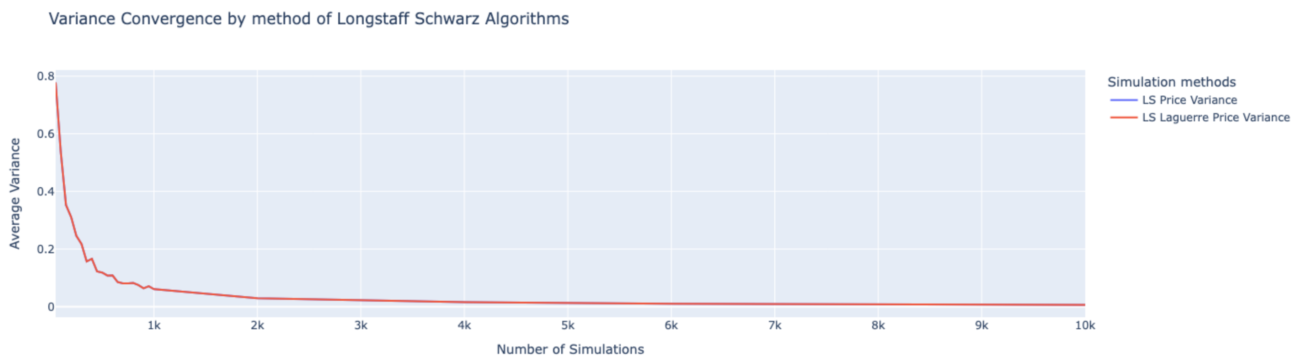

Figure 12: Longstaff Schwarz Variance of Price for European Option on Mono Underlying

Without any surprised, our values converge to the price computed in the previous part, which matches the theoretical price of the option within the Black & Scholes model. Let us further ensure that our pricer matches the previous Monte Carlo pricing for European options on a basket of 3 stocks.

	Standard LS	Laguerre LS
Price	6.743	6.743
Variance	0.081	0.081
CI	[6.585 ; 6.901]	[6.585 ; 6.901]

Table 5: LS and LS Laguerre for European Mono Underlying Call Option

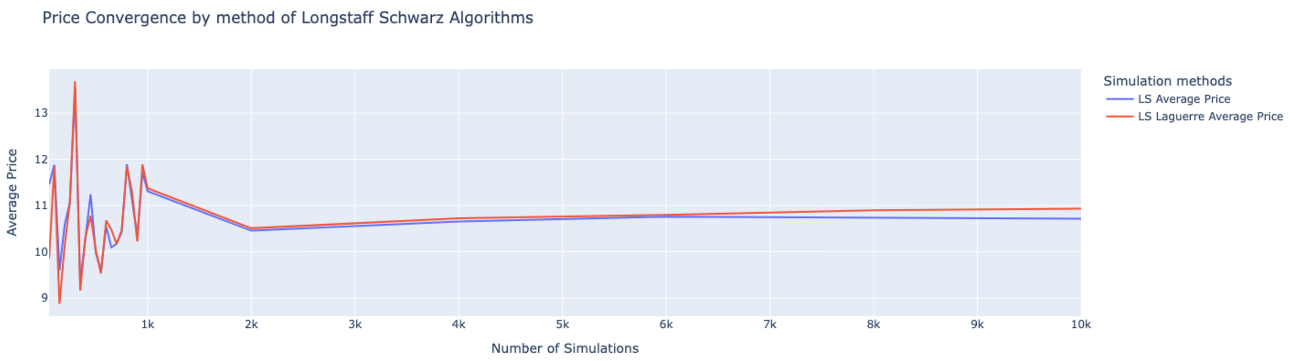

Figure 13: Longstaff Schwarz Pricing for European Option on Basket

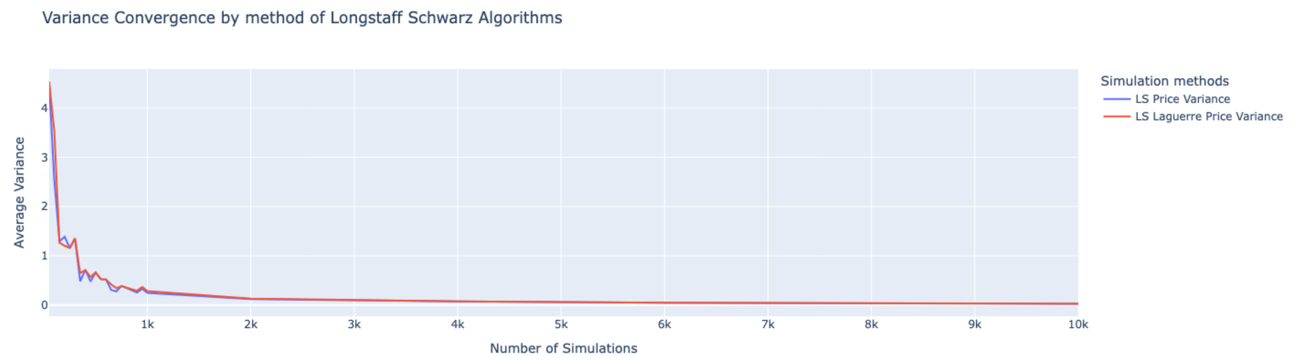

Figure 14: Longstaff Schwarz Variance of Price for European Basket

	Standard LS	Laguerre LS
Price	10.426	10.467
Variance	0.389	0.386
CI	[9.663 ; 11.188]	[9.710 ; 11.223]

Table 6: LS and LS Laguerre for European Basket Call Option

Once again, our prices agree with the values previously computed.

Overall this is reassuring given the method used here. Defining only one observation at maturity boils down to diffusing the underlying asset, computing the average value of the payoffs at maturity and ... that's all. Since we do not have any other observation dates to evaluate, the optimal strategy (optimal stopping time τ^*) is the maturity T. Thus yielding the same results

	LS	LS Antithetic	LS Control Variate	LS Combined
Price	10.467	10.776	11.048	11.195
Variance	0.386	0.339	0.018	0.034
CI	[9.710 ; 11.223]	[10.111 ; 11.442]	[11.012 ; 11.084]	[11.129 ; 11.261]

Table 7: LS Variance Reduction Techniques for European Basket Option

as Monte Carlo pricings.

It is interesting to see the variance of the different methods proposed in the previous part. We shall then see if the variance reduction techniques remain relevant for the pricing of Bermudan Options.

Longstaff Schwarz Variance Reduction for Bermudan Options Pricing

In this section, we try to evaluate the Variance Reduction capacity of the following methods:

- Antithetic Variables
- Control Variate
- Quasi Monte Carlo using Halton sequences

In this example we consider a Call option with maturity $T = 1$ and the exercise dates are the following:

$$\{0.2, 0.4, 0.6, 1.0\}$$

In order to assess our models prices we consider the following elements:

- Given the amount of "extra" optionality given by the US, it could be intuitive to think of the European Call option as a lower bound for the price of the American option. In the Black & Scholes framework, without dividends, we know that the price of the European option and American option should be equal by AOA. As we will see, the question of lower bound is not always satisfied in the simulations, which can be explained by the numerical precision of the programs and the relatively low number of simulation chosen here.
- In the context of higher and higher interest rates, we expect the difference between the price of the European At-The-Money forward put option and that of the Bermudan option with the same strike to get widen as the interest rate grows.

We remain on the basket option evaluated above and investigate the potential of Variance Reduction offered by those different techniques.

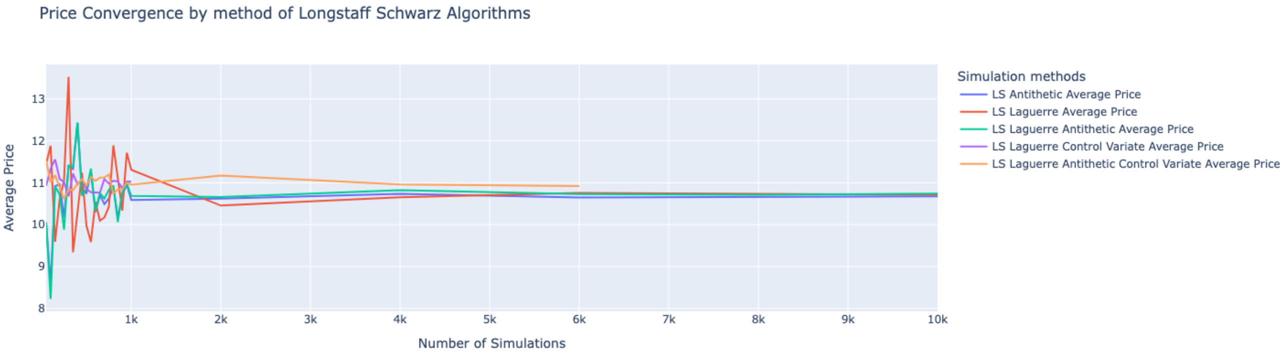

Figure 15: LS and LS Laguerre Bermudan Basket Call Option Prices

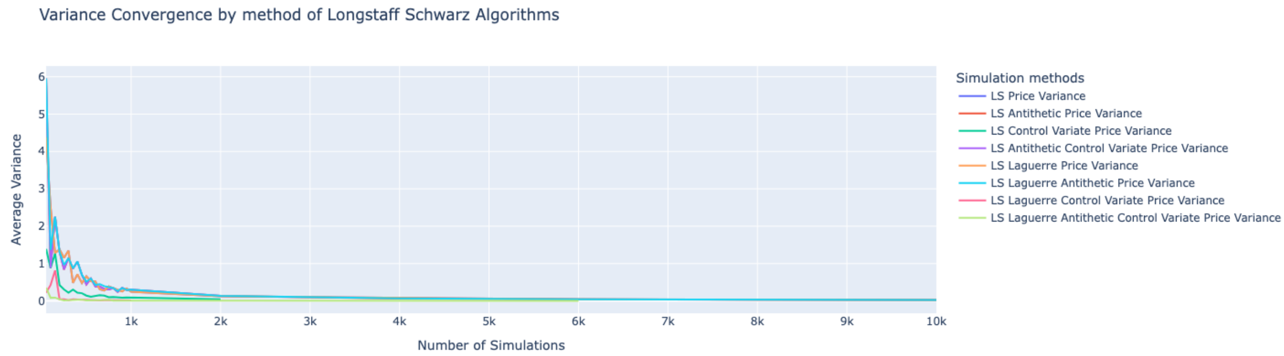

Figure 16: LS and LS Laguerre Variance Reduction for Call Bermudan Basket Option

	LS	**LS Antithetic**	**LS Control Variate**	**LS Combined**
Price	10.426	10.621	14.863	16.158
Variance	0.389	0.311	0.101	0.144
CI	[9.663 ; 11.189]	[10.011 ; 11.231]	[14.664 ; 15.062]	[15.876 ; 16.440]

Table 8: LS Variance Reduction Techniques Summary Statistics

	LS	**LS Antithetic**	**LS Control Variate**	**LS Combined**
Price	10.426	10.831	10.980	11.195
Variance	0.389	0.369	0.014	0.034
CI	[9.663 ; 11.188]	[10.108 ; 11.554]	[10.951 ; 11.006]	[11.129 ; 11.261]

Table 9: LS Laguerre Variance Reduction Techniques Summary Statistics

In both cases, we notice that variance reduction techniques indeed contribute to decreasing the variance. It is interesting to see that in our case, the most effective techniques are the techniques involving Control Variate, which work even better for the Laguerre polynomials approach. This result remains to be put into perspective given the suspicious values of the prices obtains using the Standard LS approach with the same variance reduction techniques. Additionally, we notice that thse are also the methods that converge towards 0 (see curves in Yellow, Pink and Green in figure 9.

Finally, let us see how Quasi Monte Carlo methods can be applied here. We already noted that given some proporties of the low-discrepancy sequences, combined with the method used to convert the [0, 1]-valued elements into Gaussian values, the averages were not coinciding resulting in huge differences in prices. Neverthless, let us see whether QMC methods can lead to variance reduction in the context of Bermudan Basket Options.

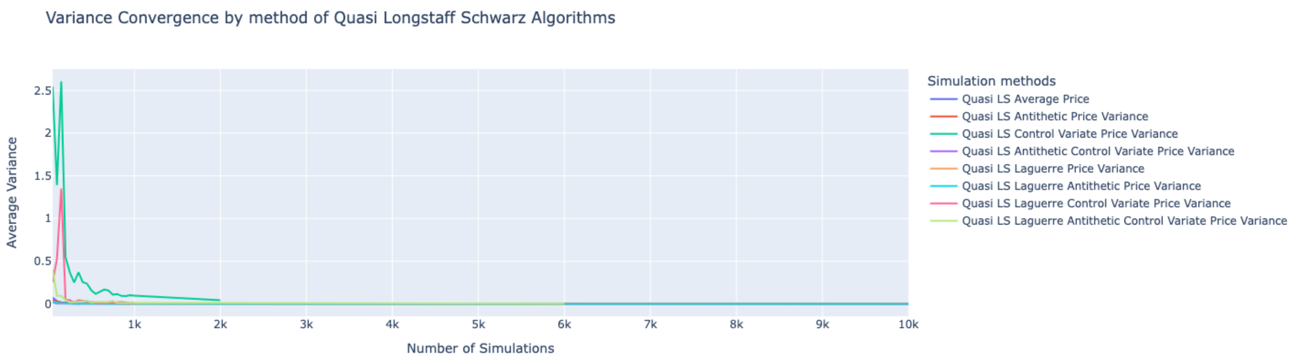

Figure 17: QuasiLS and Quasi LS Laguerre Variance Reduction for Call Bermudan Basket Option

	LS	**LS Antithetic**	**LS Control Variate**	**LS Combined**
mean	2.216	1.483	14.905	9.908
var	0.004	0.003	0.108	0.001
ci_interval	[2.207 ; 2.224]	[1.477 ; 1.489]	[14.694 ; 15.117]	[9.906 ; 9.910]

Table 10: Quasi LS Variance Reduction Techniques Summary Statistics

	LS	**LS Antithetic**	**LS Control Variate**	**LS Combined**
mean	0.823	0.878	10.949	11.184
var	0.001	0.002	0.014	0.034
ci_interval	[0.820 ; 0.826]	[0.875 ; 0.881]	[10.921 ; 10.978]	[11.117 ; 11.252]

Table 11: Quasi Laguerre LS Variance Reduction Techniques Summary Statistics

Whether it is graphically, or statistically speaking, there is no doubt that the Quasi Monte Carlo methods lead to the most significant variance reduction. We nevertheless keep in mind the huge discrepancies in terms of pricing, pushing us to consider alternative methods to price such options.

Conclusion

This project aimed to price European and Bermudan basket options using Monte Carlo and Longstaff Schwarz methods, while evaluating the impact of several variance reduction techniques.

We validated our implementations by comparing results with the Black&Scholes formula for European options. Among all techniques, Control Variate consistently provided the greatest variance reduction, especially when combined with Antithetic sampling. Laguerre polynomials also improved regression performance in the Longstaff Schwarz framework.

Quasi-Monte Carlo methods delivered significant variance reduction but introduced some inconsistencies in pricing, warranting further investigation.

Overall, variance reduction techniques substantially enhanced the efficiency and accuracy of our simulations, particularly for complex products like Bermudan options.

Appendix

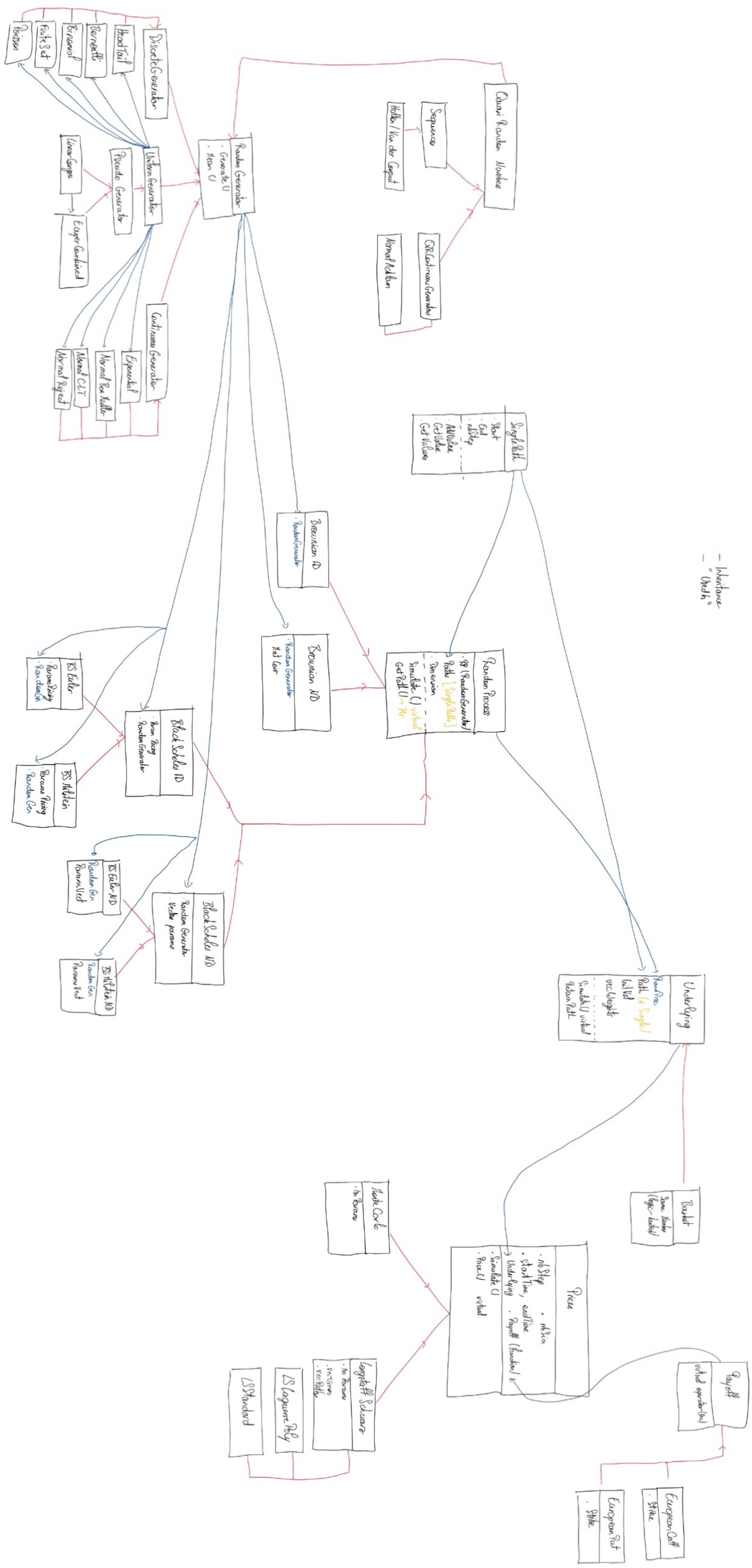

Figure 18: Program Architecture (Zoom for better clarity)